The standard CH employment-density index

Controls: Density measure—Ciccone–Hall index; CH specification—Standard CH; State (Ciccone–Hall).

1. Why use a county-based index? Two regions can have the same average density but different concentrations of employment across counties. The CH index retains this information. Let c denote a county and g an MSA or historical state/territory group. State groups use the dataset's County FE assignments for the territory year. County employment is n_c, land area is a_c, and density is $d_c = n_c/a_c$. Total employment is $N_g = \sum_{c \in g} n_c$. The index is

$$D_g(\theta) = \frac{1}{N_g} \sum_{c \in g} n_c^{\theta} a_c^{1-\theta} = \sum_{c \in g} \frac{n_c}{N_g} d_c^{\theta-1}.$$

Each county's density contribution is weighted by its employment share. The index depends on the unknown parameter θ, which the regression estimates.

2. The regression and reported coefficient. Write y_g for the selected log GDP per employment outcome, after any chosen outcome adjustments. With all industries and no adjustment, $y_g = \log(Q_g/N_g)$, where Q_g is total GDP. The model is

$$y_g = \alpha + \log D_g(\theta) + W_g'\delta + u_g.$$

Here W_g contains any selected regional controls and fixed effects, δ their coefficients, and u_g the error. The reported elasticity is $\rho = \theta - 1$, not θ. If all constituent county densities rose by 1% while their employment shares stayed fixed, the model predicts approximately ρ% higher GDP per employment. At $\rho = 0$, the standard index is one and its logarithm is zero. For one county, $\log D_g(\theta) = \rho \log d_c$, giving the ordinary county density equation.

3. How it is estimated. OLS selects θ by nonlinear least squares. IV uses the selected historical instrument in nonlinear GMM. A separate first-stage regression is unavailable for CH; the instrument-relevance statistic uses the fitted index's response to θ. IV interpretation still requires a relevant instrument and a valid exclusion restriction. If several nonlinear solutions exist, the app reports a warning and selects the solution nearest the nonlinear least-squares estimate.

4. Coverage and available settings. MSA CH permits regional controls, state fixed effects and separate outcome adjustments for schooling and college. Standard State CH uses all-industry GDP per employment, robust errors across historical regions, and no human-capital adjustment or regional fixed effects. A separate effect for every region would absorb all variation. Both models require complete county employment and land area; combined BEA units remain together. Zero-employment counties contribute zero to the index. State outcomes additionally require complete county GDP.